National College of Ireland

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MS in Cloud Computing– Full Time– Year 1–MSCCLOUD1JAN24A
MS in Cloud Computing– Full Time– Year 1– MSCLOUD_JAN24B_I

Semester Two Examinations- 2023/2024

Thursday 2nd May 2024

10.00am – 12.00pm

Blockchain Concepts and Technologies

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Duration of exam: 2 Hours
Attachments: N/A

Weighting: 50%

Answer All Questions

- Please ensure that your Handwriting is legible and accurately structured to ensure effective communication of your responses.
- Please attempt to answer the questions in Order.
- Ensure to review all questions before attempting to answer.
- Ensure that you are answering the question being asked.
- 12 Questions in Total – Two 25 Mark Questions / Ten 5 Mark Questions

- Be aware of the weighting of each question and the expected complexity and length of response.

Question 1: (5 marks)

The European Union in April of 2023 has passed legislation relating to the mining and holding of Crypto-Assets. Provide a brief summary of this legislation.

Question 2: (5 marks)

Detail and explain why you are expected to wait for a certain number of confirmations after receiving a cryptocurrency transaction?

Question 3: (5 marks)

Explain the concept of an ICO, within your response highlight a positive and negative impact of such an action.

Question 4: (5 marks)

Transactions performed on the Bitcoin and Ethereum Blockchain Networks are considered anonymous and untraceable. Is this statement considered true or false? Defend your response.

Question 5: (5 marks)

Game Theory is considered as a study of strategic decision making, initially proposed in 1949 by John Von Neumann and Oskar Morgenstern.

Provide 2 examples of such game theories and briefly discuss how they can be considered within the context of Blockchain Networks.

Question 6: (5 marks)

Explain the concept of an Alt-Coin and Briefly discuss 2 such Alt-Coins of your own choosing, highlighting the novelty and function of the chosen coin.

Question 7: (5 marks)

What prominent Ethereum based technology could possibly be considered as "A Revolution in Property Rights".

Question 8: (5 Marks)

Explain the concept of the Hyper-Ledger Fabric DLT (Distributed Ledger Technology) specifically give insights into how this Hyper-Ledger technology differs from Bitcoin and Ethereum, additionally provide a real-world example of a Hyper-ledger Fabric implementation.

Question 9 (5 marks)

The implementation of Web3 technologies and corresponding Central Bank Backed Digital currency is expected to be the next technological evolution over the coming 5 years. Provide 2 clearly defined positives and 2 negatives of such evolution.

Question 10 (5 marks)

Scalability is considered a persistent issue in terms of growth and capability of Both the Bitcoin Blockchain and the Ethereum Blockchain respectively.
Give brief insights into the negative impact of scalability in BTC and ETH blockchain networks.
Additionally give 2 brief examples of measures which have been implemented to help to resolve scalability issues in the blockchain networks mentioned.

Question 11 (25 marks)

Blockchain is considered as a sub-set of Distributed Ledger Technologies (DLT) which is often referred to as a paradigm ‘a combination of varied technologies’.

Explain and discuss in detail the technological advancements which you believe have enabled the creation of Bitcoin in 2009. Within your response ensure to:

- Give reference to key contributors of the chosen discussed technologies.
- Ensure to provide insights as to how these technological contributions have been implemented within the Bitcoin blockchain.

Question 12 (25 marks)

Achieving consensus is a fundamental requirement of a Blockchain Network.

Explain in detail the Proof of Work (POW) consensus mechanism used in the Bitcoin Blockchain. Points to consider as part of your response.

- Ensure to provide a fine level of granularity in your technical explanation of POW.
- A Diagram can be used to further the clarity of your response.
- Discuss how POW differs from the Proof of Stake (POS) consensus mechanism.
- Highlight and discuss recent events relating to POW.